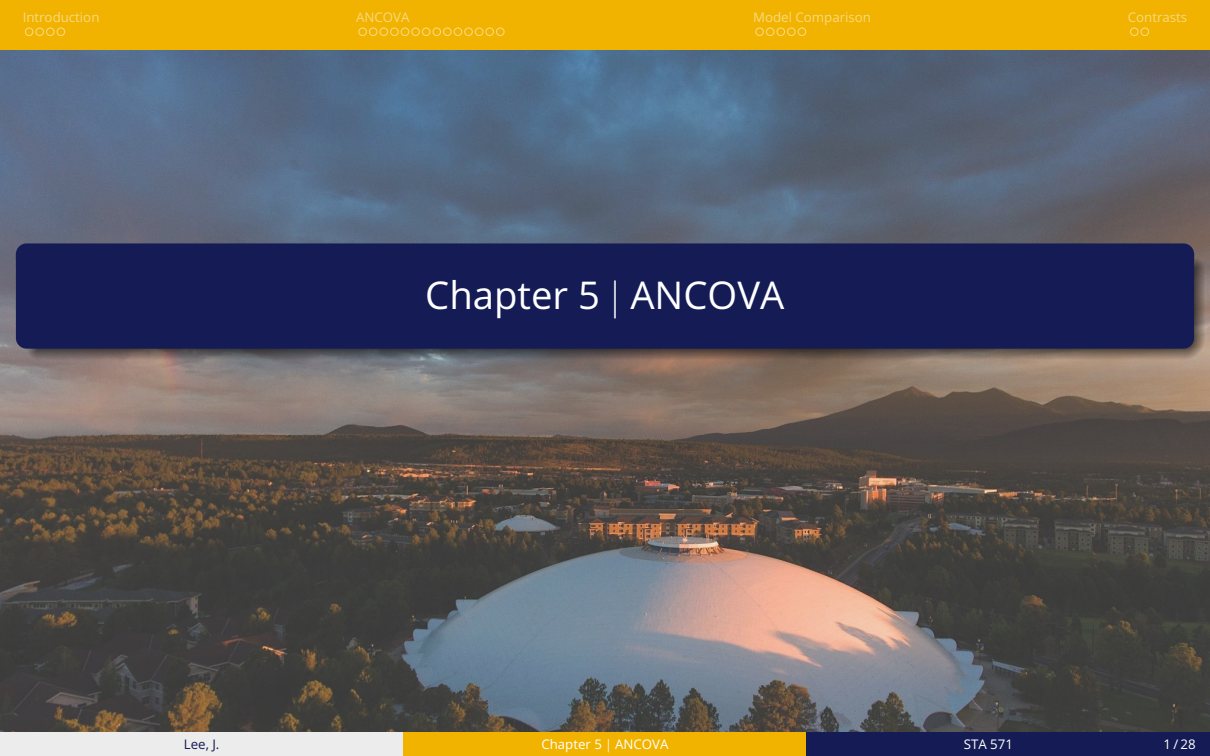
Chapter 5 | ANCOVA



Learning objectives

- Understand the concept of Analysis of Covariance (ANCOVA) model.
- Visualize the relation between response variable with its predictors.
- Fit ANCOVA models and use statistical tests (*t*-test and *F*-test) to determine an optimal ANCOVA model.
- Interpret the relation between the response variable with its predictors in the context of the problem.
- Construct a 95% confidence interval and use a *t*-test for a contrast $\mathbf{c}^{\mathrm{T}}\boldsymbol{\beta}$ in ANCOVA.

Outline

1. Introduction
2. ANCOVA
3. Model Comparison
4. Contrasts

Building blocks | SLR

- Simple linear regression (SLR): $Y_i = \beta_0 + \beta_1 x_i + \epsilon_i$ for $i = 1, \ldots, n$.
  1. The response Y is a continuous variable to be observed from each subject.
  2. The predictor X is a continuous or numerical variable to be observed from each subject.
  3. Parameter of main interest: β_1. Because $E(Y|x) = \beta_0 + \beta_1 x$, we estimate, for every 1-unit increase in predictor X, the mean response to change (increase or decrease) by $\hat{\beta}_1$.
  4. SLR is a linear model (LM):

$$\begin{cases} Y_1 = \beta_0 + \beta_1 x_1 + \epsilon_1 \\ \vdots \\ Y_n = \beta_0 + \beta_1 x_n + \epsilon_n \end{cases} \Rightarrow \begin{bmatrix} Y_1 \\ \vdots \\ Y_n \end{bmatrix} = \beta_0 \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} + \beta_1 \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} \epsilon_1 \\ \vdots \\ \epsilon_n \end{bmatrix} \Rightarrow \begin{bmatrix} Y_1 \\ \vdots \\ Y_n \end{bmatrix} = \begin{bmatrix} 1 & x_1 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} + \begin{bmatrix} \epsilon_1 \\ \vdots \\ \epsilon_n \end{bmatrix}$$

Therefore, SLR has a unified LM representation: $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$

Building blocks | One-way ANOVA

- One-way offset ANOVA: $Y_{ij} = \mu + \tau_i + \epsilon_{ij}$ for $i = 1, \ldots, m$; $j = 1, \ldots, n_i$.
  1. The response Y is a continuous variable to be observed from subject j within group i.
  2. Parameters of main interest: $\tau_2, \ldots, \tau_m$. Because $E(Y_{ij}) - E(Y_{1j}) = (\mu + \tau_i) - \mu = \tau_i$, we estimate the difference in the mean response from group i to group 1 is $\hat{\tau}_i$; the mean response of group i is greater or smaller than group 1 by $\hat{\tau}_i$.
  3. One-way ANOVA is a linear model (LM). Suppose $m = 2$ and $n_1 = n_2 = 5$. Then,

$$\begin{aligned} Y_{11} &= \mu + \epsilon_{11} \\ &\vdots \\ Y_{15} &= \mu + \epsilon_{15} \\ Y_{21} &= \mu + \tau_2 + \epsilon_{21} \\ &\vdots \\ Y_{25} &= \mu + \tau_2 + \epsilon_{25} \end{aligned} \Rightarrow \begin{bmatrix} Y_{11} \\ \vdots \\ Y_{15} \\ Y_{21} \\ \vdots \\ Y_{25} \end{bmatrix} = \mu \begin{bmatrix} 1 \\ \vdots \\ 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} + \tau_2 \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ \vdots \\ 1 \end{bmatrix} + \begin{bmatrix} \epsilon_{11} \\ \vdots \\ \epsilon_{15} \\ \epsilon_{21} \\ \vdots \\ \epsilon_{25} \end{bmatrix} \Rightarrow \begin{bmatrix} Y_{11} \\ \vdots \\ Y_{15} \\ Y_{21} \\ \vdots \\ Y_{25} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ \vdots & \vdots \\ 1 & 0 \\ 1 & 1 \\ \vdots & \vdots \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \mu \\ \tau_2 \end{bmatrix} + \begin{bmatrix} \epsilon_{11} \\ \vdots \\ \epsilon_{15} \\ \epsilon_{21} \\ \vdots \\ \epsilon_{25} \end{bmatrix}$$

Like SLR, one-way ANOVA has a unified LM representation: $\mathbf{y} = \mathbf{X}\beta + \varepsilon$

Building blocks | One-way ANOVA

- One-way offset ANOVA:
  4. No predictor? In fact, ANOVA contains a "hidden" predictor. Let $\mathbf{x}$ denote the 2nd column of the ANOVA design matrix $\mathbf{X}$ and reexpress as:

$$\mathbf{x} = \mathbf{X}_{[,2]} = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ \vdots \\ 1 \end{bmatrix} = \begin{bmatrix} x_{11} \\ \vdots \\ x_{15} \\ x_{21} \\ \vdots \\ x_{25} \end{bmatrix}, \quad x_{ij} = \begin{cases} 0, & \text{if subject } j \text{ is in group 1;} \\ 1, & \text{if subject } j \text{ is in group 2} \end{cases}$$

     Because we know which group subject j is included, x_{ij} is given or observed.
  5. Therefore, ANOVA contains the predictor x_{ij} that is called a **factor** (or a categorical variable).
  6. Caution: Unless the level of factor x is 2, we should not treat factor as a numerical variable.[1]

[1] The factor x indicates which group the subject is included. The difference between $x = 3$ (group 3) and $x = 2$ (group 2) is not necessarily meant to be the same as the difference between $x = 2$ (group 2) and $x = 1$ (group 1). If there are $m = 3$ groups, we need a total of $m - 1 = 2$ indicator variables to specify the three groups: $x_2 = 1$ if subject j is in group 2, and 0 otherwise; and $x_3 = 1$ if subject j is in group 3, and 0 otherwise

Combining building blocks | ANCOVA

- In summary,

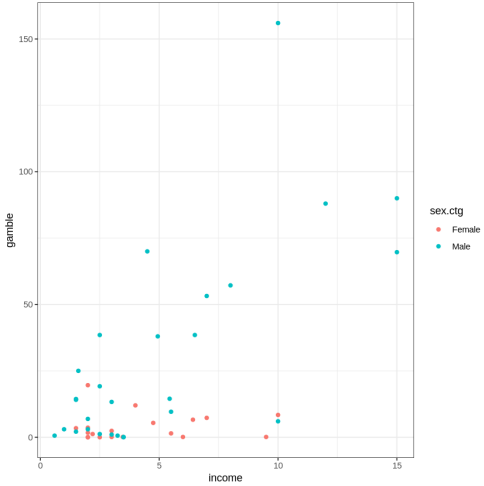
Statistical model	Response Y	Predictor X
SLR	continuous	continuous (or numerical)
ANOVA	continuous	factor (or categorical variable)

- Suppose the data contains two predictors X_1 and X_2 and continuous response Y. If one of the predictors is continuous and the other is a factor (or a categorical variable), then **one-way Analysis of Covariance (ANCOVA)** can be used.

Example | One-way ANCOVA

- Example 5.1. The dataset `teengamb` in the `faraway` package contains data regarding the rates of gambling among teenagers in Britain and their gender and socioeconomic status. We are interested in how gender and income relate to how much a person gambles.

	sex	income	gamble
1	1	2.00	0.0
2	1	2.50	0.0
3	1	2.00	0.0
4	1	7.00	7.3
5	1	2.00	19.6
6	1	3.47	0.1
:	:	:	:



Three models

- We consider the following three models.
  1. **Model 1 (SLR)**: The relation between income and gambling rate is the same for each gender.
  2. **Model 2 (Additive model with offset parallel lines)**: The mean level of gambling rate is different for each gender. However, the relation between income and gambling rate remains the same for each gender.
  3. **Model 3 (Interaction model with nonparallel lines)**: The mean level of gambling rate is different for each gender. In addition, the effect of income on gambling rate is different for each gender.

One-way ANCOVA | Model 1 (SLR)

- **Model 1 (SLR)**: The relation between income X and gambling rate Y is the same for each gender.
  1. If there are no group effects on the response Y, then the y-intercepts for each group are the same and also the slopes for each group are the same.
  2. We model (x_i, y_i) as a simple linear regression: $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$ for $i = 1, \ldots, n$
  3. Model 1 (SLR) has the following linear model representations:

$$\begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} = \beta_0 \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} + \beta_1 \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} \epsilon_1 \\ \vdots \\ \epsilon_n \end{bmatrix} \Rightarrow \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} 1 & x_1 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} + \begin{bmatrix} \epsilon_1 \\ \vdots \\ \epsilon_n \end{bmatrix} \Rightarrow \mathbf{y} = \mathbf{X}^{(1)} \boldsymbol{\beta}^{(1)} + \boldsymbol{\epsilon}$$

  4. The design matrix and parameter vector for Model 1 are:

$$\mathbf{X}^{(1)} = \begin{bmatrix} 1 & x_1 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix} \text{ and } \boldsymbol{\beta}^{(1)} = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}$$

One-way ANCOVA | Model 1 (SLR)

- **Model 1 (SLR)**:
  5. We fit Model 1 by using `lm()` function:

```
> m1 <- lm( gamble ~ income, data=df.gamb )
> summary( m1 )

             Estimate Std. Error t value Pr(>|t|)
(Intercept)    -6.325      6.030  -1.049      0.3
income          5.520      1.036   5.330 3.05e-06 ***

F-statistic: 28.41 on 1 and 45 DF,  p-value: 3.045e-06
```

  6. The estimated linear line is

$$\hat{\mu}_{Y|x} = \hat{\mathrm{E}}(Y|x) = \hat{\beta}_0 + \hat{\beta}_1 x = -6.325 + 5.520x$$

  7. Both t-test ($t = 5.330$) and F-test ($F = 28.41$) reject $H_0 : \beta_1 = 0$ with identical p-value (= 0.000003), because $t^2 = F$ (i.e., $5.330^2 = 28.41$).

One-way ANCOVA | Model 2 (Additive model)

- **Model 2 (Additive model)**: The mean level of gambling rate Y is different for each gender. However, the relation between income X and gambling rate Y remains the same for each gender.
  1. The y-intercepts for each group are different but the slopes are the same for each group.
  2. We model the data (x_i, y_i) as follows:

$$y_i = \begin{cases} \beta_0^{\mathrm{I}} + \beta_1 x_i + \epsilon_i, & \text{if } i \in \text{group I} \\ \beta_0^{\mathrm{II}} + \beta_1 x_i + \epsilon_i, & \text{if } i \in \text{group II} \end{cases} \Rightarrow y_i = \begin{cases} \beta_0 + \beta_1 x_i + \epsilon_i, & \text{if } i \in \text{group I} \\ (\beta_0 + \beta_0^*) + \beta_1 x_i + \epsilon_i, & \text{if } i \in \text{group II} \end{cases}$$

  3. This can be simplified to a single equation by using an indicator (dummy) variable $1_{[i \in \text{group II}]}$:

$$y_i = \beta_0 + \beta_0^* 1_{[i \in \text{group II}]} + \beta_1 x_i + \epsilon_i, \quad 1_{[i \in \text{group II}]} = \begin{cases} 1, & \text{if } i \in \text{group II}; \\ 0, & \text{if not.} \end{cases}$$

  4. Interpretation of β_0^*: Because

$$E(Y|x, \text{group II}) - E(Y|x, \text{group I}) = (\beta_0 + \beta_0^* + \beta_1 x) - (\beta_0 + \beta_1 x) = \beta_0^*,$$

  β_0^* represents the difference in the mean response between groups I and II at the same value of predictor X.

One-way ANCOVA | Model 2 (Additive model)

- **Model 2 (Additive model)**:
  5. This model is known as an **additive model with offset parallel lines**.
  6. If the first n_1 pairs of (x_i, y_i) are in group I and the other n_2 pairs are in group II, then

$$\begin{bmatrix} y_1 \\ \vdots \\ y_{n_1} \\ y_{n_1+1} \\ \vdots \\ y_{n_1+n_2} \end{bmatrix} = \beta_0 \begin{bmatrix} 1 \\ \vdots \\ 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} + \beta_0^* \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ \vdots \\ 1 \end{bmatrix} + \beta_1 \begin{bmatrix} x_1 \\ \vdots \\ x_{n_1} \\ x_{n_1+1} \\ \vdots \\ x_{n_1+n_2} \end{bmatrix} + \begin{bmatrix} \epsilon_1 \\ \vdots \\ \epsilon_{n_1} \\ \epsilon_{n_1+1} \\ \vdots \\ \epsilon_{n_1+n_2} \end{bmatrix} \Rightarrow \begin{bmatrix} y_1 \\ \vdots \\ y_{n_1} \\ y_{n_1+1} \\ \vdots \\ y_{n_1+n_2} \end{bmatrix} = \begin{bmatrix} 1 & 0 & x_1 \\ \vdots & \vdots & \vdots \\ 1 & 0 & x_{n_1} \\ 1 & 1 & x_{n_1+1} \\ \vdots & \vdots & \vdots \\ 1 & 1 & x_{n_1+n_2} \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_0^* \\ \beta_1 \end{bmatrix} + \begin{bmatrix} \epsilon_1 \\ \vdots \\ \epsilon_{n_1} \\ \epsilon_{n_1+1} \\ \vdots \\ \epsilon_{n_1+n_2} \end{bmatrix}$$

One-way ANCOVA | Model 2 (Additive model)

- **Model 2 (Additive model)**:
  7. Model 2 has a unified LM representation:

$$\boxed{\mathbf{y} = \mathbf{X}^{(2)}\boldsymbol{\beta}^{(2)} + \boldsymbol{\epsilon}},$$

where the design matrix and parameter vector for Model 2 are:

$$\mathbf{X}^{(2)} = \begin{bmatrix} 1 & 0 & x_1 \\ \vdots & \vdots & \vdots \\ 1 & 0 & x_{n_1} \\ 1 & 1 & x_{n_1+1} \\ \vdots & \vdots & \vdots \\ 1 & 1 & x_{n_1+n_2} \end{bmatrix} \text{ and } \boldsymbol{\beta}^{(2)} = \begin{bmatrix} \beta_0 \\ \beta_0^* \\ \beta_1 \end{bmatrix}$$

  8. Note: If $\beta_0^* = 0$, then $\mathbf{X}^{(2)} = \mathbf{X}^{(1)}$ and $\boldsymbol{\beta}^{(2)} = \boldsymbol{\beta}^{(1)}$, and therefore, the model becomes Model 1. That is, Model 2 with $\beta_0^* = 0$ is Model 1.

One-way ANCOVA | Model 2 (Additive model)

- **Model 2 (Additive model)**:
  8. We fit Model 2 by using `lm()` function:

```
> df.gamb <- teengamb |>
>              mutate( sex.ctg = ifelse( sex==1, 'Female', 'Male' ) )
> m2 <- lm( gamble ~ sex.ctg + income, data=df.gamb )
> summary( m2 )

             Estimate Std. Error t value Pr(>|t|)
(Intercept)  -17.594      6.544  -2.688  0.01010 *
sex.ctgMale   21.634      6.809   3.177  0.00272 **
income         5.172      0.951   5.438 2.24e-06 ***

F-statistic: 22.12 on 2 and 44 DF,  p-value: 2.243e-07
```

  9. The unified linear line is estimated by

$$\hat{\mu}_{Y|x} = \hat{\mathrm{E}}(Y|x) = \hat{\beta}_0 + \hat{\beta}_0^* \cdot 1_{[\text{group II}]} + \hat{\beta}_1 x = -17.594 + 21.634 \cdot 1_{[\text{group II}]} + 5.172x$$

  10. t-test ($t = 3.177$) rejects $H_0 : \beta_0^* = 0$ with p-value = 0.00272.

One-way ANCOVA | Model 3 (Interaction model)

- **Model 3 (Interaction model)**: The mean level of gambling rate Y is different for each gender. Also, the effect of income X on gambling rate Y is different for each gender.
  1. The y-intercepts are different for each group, and the slopes are also different for each group.
  2. We model the data (x_i, y_i) as follows:

$$y_i = \begin{cases} \beta_0^{\mathrm{I}} + \beta_1^{\mathrm{I}} x_i + \epsilon_i, & \text{if } i \in \text{group I} \\ \beta_0^{\mathrm{II}} + \beta_1^{\mathrm{II}} x_i + \epsilon_i, & \text{if } i \in \text{group II} \end{cases} \Rightarrow y_i = \begin{cases} \beta_0 + \beta_1 x_i + \epsilon_i, & \text{if } i \in \text{group I} \\ (\beta_0 + \beta_0^*) + (\beta_1 + \beta_1^*) x_i + \epsilon_i, & \text{if } i \in \text{group II} \end{cases}$$

  3. This can be simplified to a single equation by using an indicator (dummy) variable $1_{[i \in \text{group II}]}$:

$$y_i = \beta_0 + \beta_0^* 1_{[i \in \text{group II}]} + (\beta_1 + \beta_1^* 1_{[i \in \text{group II}]}) x_i + \epsilon_i, \quad 1_{[i \in \text{group II}]} = \begin{cases} 1, & \text{if } i \in \text{group II}; \\ 0, & \text{if not.} \end{cases}$$

  4. This model is known as an **interaction model (separate lines)**.

One-way ANCOVA | Model 3 (Interaction model)

- **Model 3 (Interaction model)**:
  5. Interpretation of β_1^*: As the predictor X increases by 1-unit (from x to $x+1$), change of the mean response at each group is

$$\begin{aligned} E(Y|x+1, \text{group I}) - E(Y|x, \text{group I}) &= [\beta_0 + \beta_1(x+1)] - [\beta_0 + \beta_1 x] = \beta_1 \\ E(Y|x+1, \text{group II}) - E(Y|x, \text{group II}) &= [\beta_0 + \beta_0^* + (\beta_1 + \beta_1^*)(x+1)] - [\beta_0 + \beta_0^* + (\beta_1 + \beta_1^*)x] \\ &= \beta_1 + \beta_1^* \end{aligned}$$

After subtracting these changes, for every 1-unit increase in X, we have

$$\begin{aligned} & [E(Y|x+1, \text{group II}) - E(Y|x, \text{group II})] - [E(Y|x+1, \text{group I}) - E(Y|x, \text{group I})] = \beta_1^* \\ \Rightarrow\ & [E(Y|x+1, \text{group II}) - E(Y|x, \text{group II})] = [E(Y|x+1, \text{group I}) - E(Y|x, \text{group I})] + \beta_1^* \\ \Rightarrow\ & \boxed{\text{Change of mean response in group II} = \text{Change of mean response in group I} + \beta_1^*} \end{aligned}$$

One-way ANCOVA | Model 3 (Interaction model)

- **Model 3 (Interaction model)**:
  6. We model the data (x_i, y_i) as follows:

$$\begin{bmatrix} y_1 \\ \vdots \\ y_{n_1} \\ y_{n_1+1} \\ \vdots \\ y_{n_1+n_2} \end{bmatrix} = \beta_0 \begin{bmatrix} 1 \\ \vdots \\ 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} + \beta_0^* \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ \vdots \\ 1 \end{bmatrix} + \beta_1 \begin{bmatrix} x_1 \\ \vdots \\ x_{n_1} \\ x_{n_1+1} \\ \vdots \\ x_{n_1+n_2} \end{bmatrix} + \beta_1^* \begin{bmatrix} 0 \\ \vdots \\ 0 \\ x_{n_1+1} \\ \vdots \\ x_{n_1+n_2} \end{bmatrix} + \begin{bmatrix} \epsilon_1 \\ \vdots \\ \epsilon_{n_1} \\ \epsilon_{n_1+1} \\ \vdots \\ \epsilon_{n_1+n_2} \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} y_1 \\ \vdots \\ y_{n_1} \\ y_{n_1+1} \\ \vdots \\ y_{n_1+n_2} \end{bmatrix} = \begin{bmatrix} 1 & 0 & x_1 & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 1 & 0 & x_{n_1} & 0 \\ 1 & 1 & x_{n_1+1} & x_{n_1+1} \\ \vdots & \vdots & \vdots & \vdots \\ 1 & 1 & x_{n_1+n_2} & x_{n_1+n_2} \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_0^* \\ \beta_1 \\ \beta_1^* \end{bmatrix} + \begin{bmatrix} \epsilon_1 \\ \vdots \\ \epsilon_{n_1} \\ \epsilon_{n_1+1} \\ \vdots \\ \epsilon_{n_1+n_2} \end{bmatrix}$$

One-way ANCOVA | Model 3 (Interaction model)

- **Model 3 (Interaction model)**:
  7. Model 3 has a unified LM representation:

$$\boxed{\mathbf{y} = \mathbf{X}^{(3)}\boldsymbol{\beta}^{(3)} + \boldsymbol{\epsilon}},$$

where the design matrix and parameter vector for Model 3 are:

$$\mathbf{X}^{(3)} = \begin{bmatrix} 1 & 0 & x_1 & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 1 & 0 & x_{n_1} & 0 \\ 1 & 1 & x_{n_1+1} & x_{n_1+1} \\ \vdots & \vdots & \vdots & \vdots \\ 1 & 1 & x_{n_1+n_2} & x_{n_1+n_2} \end{bmatrix} \text{ and } \boldsymbol{\beta}^{(3)} = \begin{bmatrix} \beta_0 \\ \beta_0^* \\ \beta_1 \\ \beta_1^* \end{bmatrix}$$

  8. Note: If $\beta_1^* = 0$, then $\mathbf{X}^{(3)} = \mathbf{X}^{(2)}$ and $\boldsymbol{\beta}^{(3)} = \boldsymbol{\beta}^{(2)}$, and therefore, the model becomes Model 2. That is, Model 3 with $\beta_1^* = 0$ is Model 2.

One-way ANCOVA | Model 3 (Interaction model)

- **Model 3 (Interaction model)**:
  9. We fit Model 3 by using `lm()` function:

```
> df.gamb <- teengamb |>
>            mutate( sex.ctg = ifelse( sex==1, 'Female', 'Male' ) )
> m3 <- lm( gamble ~ sex.ctg + income + sex.ctg:income, data=df.gamb )
> summary( m3 )

                    Estimate Std. Error t value Pr(>|t|)
(Intercept)           3.1400     9.2492   0.339  0.73590
sex.ctgMale          -5.7996    11.2003  -0.518  0.60724
income                0.1749     1.9034   0.092  0.92721
sex.ctgMale:income    6.3432     2.1446   2.958  0.00502 **

F-statistic: 20.26 on 3 and 43 DF,  p-value: 2.451e-08
```

  10. The estimated linear line is

$$\hat{\mu}_{Y|x} = \hat{\mathrm{E}}(Y|x) = 3.1400 - 5.7996 \cdot 1_{[\text{group II}]} + (0.1749 + 6.3432 \cdot 1_{[\text{group II}]})x$$

One-way ANCOVA | Model 3 (Interaction model)

- **Model 3 (Interaction model)**:
  11. t-test ($t = 2.958$) rejects $H_0 : \beta_1^* = 0$ with p-value = 0.00502, implying that the slopes are different for female and male groups.
  12. The LS estimation result from `m3` shows $\hat{\beta}_1^* = 6.3432$, meaning that the slope for males is higher than that for females by 6.3432. That is, we estimate that gambling rate among male teenagers is on average higher than female teenagers by 6.3432 for every 1-unit increase in income.

Main questions

- Three main questions are:
  1. Which of the three models would best explain the data?
  2. What are the correct interpretations for the estimated parameters of the best model?
  3. How do we inference on contrasts of interest?

Comparison of three models

- We choose an optimal model for the data. Because Model 1 is nested to Model 2 and Model 2 is nested to Model 3, an F-test can be used.
- Comparison of Model 1 and Model 2
  1. We use the `anova()` command:

```
> anova( m1, m2 )

  Res.Df        RSS   Df  Sum of Sq        F   Pr(>F)
1     45   28008.59   NA         NA       NA       NA
2     44   22781.32    1   5227.268   10.096   0.0027
```

  2. Note: Model 2 with $\beta_0^* = 0$ is Model 1.
  3. F-test ($F = 10.096$) rejects $H_0 : \beta_0^* = 0$ (Model 1) with p-value = 0.0027, which is the same result as the t-test ($t = 3.177$) for $H_0 : \beta_0^* = 0$ from the Model 2 fit result in `m2`.
  4. Interpretation: Model 2 is a highly significant improvement to Model 1.

Comparison of three models

- Comparison of Model 2 and Model 3
  1. We use the `anova()` command:

```
> anova( m2, m3 )

  Res.Df        RSS   Df  Sum of Sq         F   Pr(>F)
1     44   22781.32   NA         NA        NA       NA
2     43   18929.91    1   3851.411    8.7486   0.0050
```

  2. Note: Model 3 with $\beta_1^* = 0$ is Model 2.
  3. F-test ($F = 8.7486$) rejects $H_0 : \beta_1^* = 0$ (Model 2) with p-value = 0.0050, which is the same result as the t-test ($t = 2.958$) for $H_0 : \beta_1^* = 0$ from the Model 3 fit result in `m3`.
  4. Interpretation: Model 3 is a highly significant improvement to Model 2.

Optimal model and CI

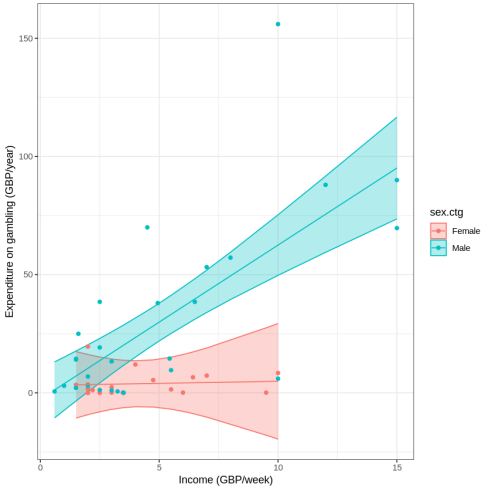
- We have determined an optimal model for the data: Model 3 Interaction model with nonparallel lines
- Optimal model and 95% CI for the mean response
  1. We use the `predict()` function:

```
> df.gamb_m3 <- df.gamb |>
>               cbind( predict( m3, interval='confidence' ) )
>
> ggplot( df.gamb_m3, aes( x=income, col=sex.ctg, fill=sex.ctg ) ) +
>   geom_ribbon( aes( ymin=lwr, ymax=upr ), alpha=.3 ) +
>   geom_point( aes( y=gamble ) ) +
>   geom_line( aes( y=fit ) ) +
>   theme_bw()
```

Interpretation

- Interpretation on important model parameter estimates:

1. Based on Model 3 (`m3`), the estimated slope for females is $\hat{\beta}_1 = 0.1749$, meaning that females spend on average 0.1749 GBP/year more for every 1 GBP/week increase in income, although this slope is not significant with a *p*-value of 0.9272.
2. The difference in the estimated slopes between males and females is $\hat{\beta}_1^* = 6.3432$, implying that males spend on average 6.3432 GBP/year more than females for 1 GBP/week increase in income. This difference is statistically significant (*p*-value = 0.0050) at $\alpha = 0.05$.



Estimation of the mean response

- Suppose we estimate the mean spending on gambling at each income level of 5 and 10 for each gender.
  1. We use the `emmeans()` function:

```
> emmeans( m3, specs=~income*sex.ctg,
>      at=list(income=c(5,10), sex.ctg=c('Male','Female')), infer=c(TRUE,TRUE) )

income  sex.ctg   emmean       SE  df  lower.CL  upper.CL  t.ratio     p.value
     5     Male  29.9309   3.9652  43   21.9343   37.9276   7.5483  2.0865e-09
    10     Male  62.5215   6.3532  43   49.7089   75.3342   9.8408  1.4017e-12
     5   Female   4.0145   5.0784  43   -6.2271   14.2562   0.7905  4.3356e-01
    10   Female   4.8891  12.1315  43  -19.5764   29.3547   0.4030  6.8893e-01
```

  2. For instance, if the income of a male is 5 GBP/week, we estimate the male spends on average 29.93 GBP/year on gambling.
  3. At the same income level, we estimate a female spends on average 4.01 GBP/year on gambling.

Estimation of the mean response difference

- Suppose we are interested in the difference of the mean responses for different genders at a certain income level.
  1. We use `emmeans()` with a `pairwise` in the `specs` argument as follows:

```
> summary( emmeans( m3, specs=pairwise~sex.ctg|income,
>    at=list(income=c(5,10), sex.ctg=c('Male','Female')), infer=c(TRUE,TRUE) ) )

$contrasts
     contrast income estimate       SE  df  lower.CL upper.CL t.ratio  p.value
Male - Female      5  25.9164   6.4431  43   12.9226  38.9102  4.0223  0.00023
Male - Female     10  57.6324  13.6945  43   30.0148  85.2500  4.2084  0.00013
```

  2. When income is 5 GBP/week, males spend on average 25.92 GBP/year more than females on gambling. With 95% confidence, the mean spending on gambling is higher for males than for females in a range of 12.92 and 38.91 GBP/year.
  3. At 10 GBP/week income level, males spend on average 57.63 GBP/year more than females on gambling, with a 95% CI of 30.01 and 85.25 GBP/year.